# DELIVERING EFFICIENCY AND SUPPORT WITH A KINETIC BUDDY(DESK)

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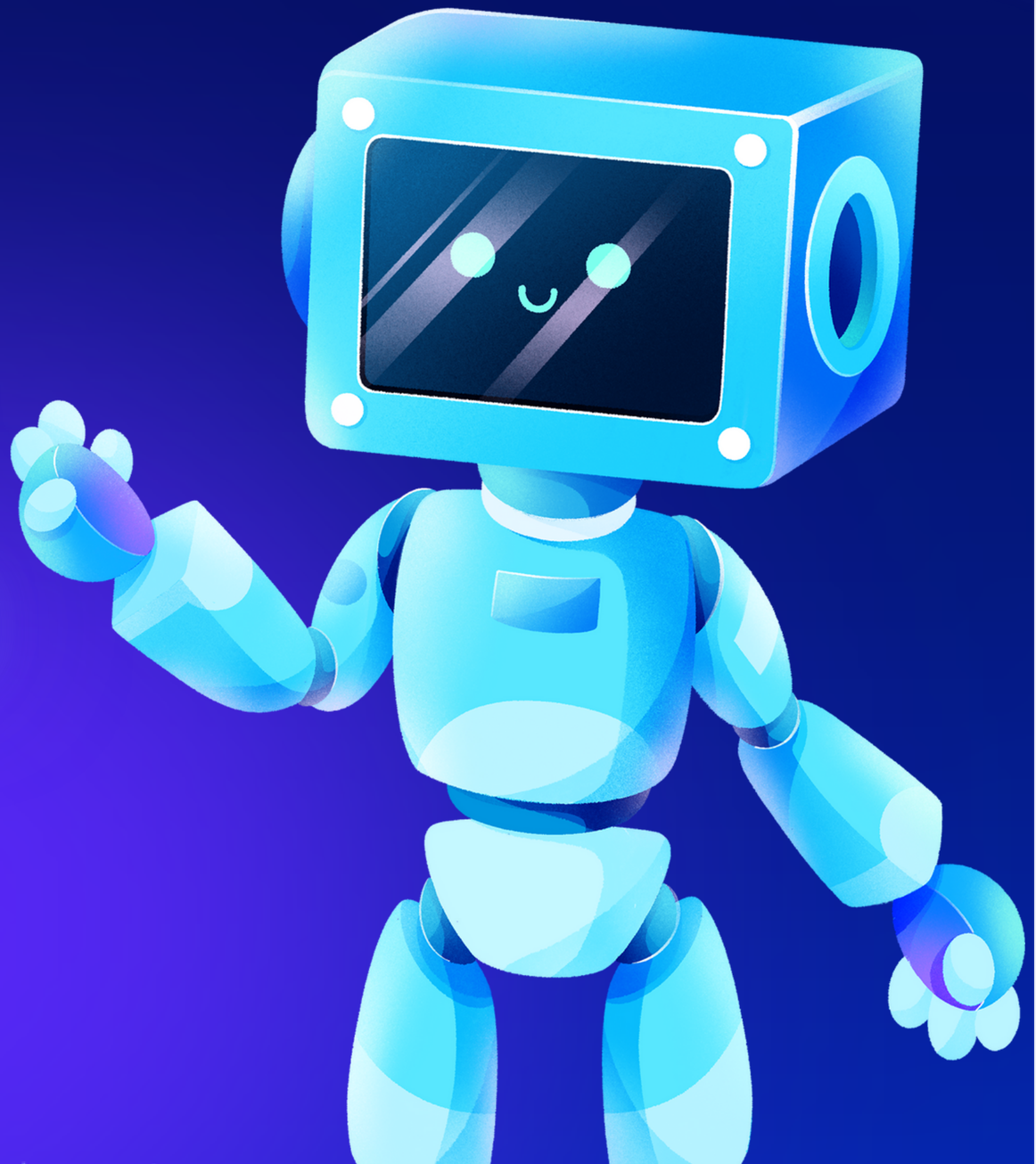
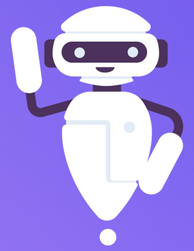


# PROBLEM STATEMENT

In today's fast-paced work and study environments, individuals often lack a supportive presence at their desks, leading to feelings of loneliness, stress, and decreased focus. Despite technological advancements, there is still a need for a tangible, interactive companion to enhance productivity and well-being. The absence of such a tool results in reduced efficiency and sub optimal performance.

To address this, there is a critical need for a "Desk Buddy" system—a personalized, interactive companion offering emotional support, motivation, task organization, and gentle break reminders. Implementing this solution would lead to improved focus, reduced stress, enhanced performance, and better mental well-being for employees and students, revolutionizing work and study habits.

# SUSTAINABILITY GOALS OF THE PROJECT

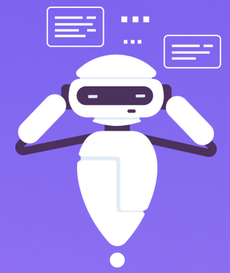


## SDG4

A smart desk study buddy can tailor its instruction to the individual needs of each student, providing them with the support they need to succeed.

## SDG8

Introducing smart desk study buddies not only boosts worker productivity but also stimulates job creation and new business ventures within the educational technology sector.



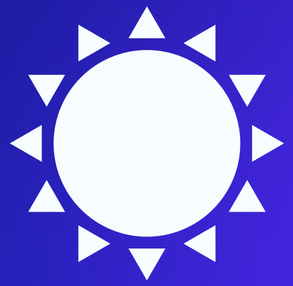
## SDG9

Smart desk study buddies promote innovative technology, stimulate educational tech industry growth, and enhance learning environments in schools and universities.

# DESK BUDDY

01 Temperature

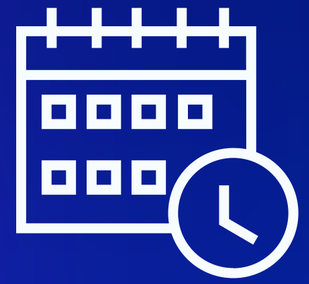
02 Humidity



03 Light Intensity

04 Air Quality

05 Calendars and Timers



# IMPACT OF DESK BUDDY

## Employment

- **Direct:** Creation of jobs in manufacturing, programming, and customer support for Desk Buddy production and maintenance.
- **Indirect:** Job creation in associated industries like supply chain, logistics, and retail.

## Economic Benefit

- **Commodity:** Increased demand for electronic components, sensors, and materials used in Desk Buddy production, stimulating related industries.
- **Service:** Growth in the market for technical support, maintenance, and software updates for Desk Buddy users, creating a service-oriented job market

## Monetary Impact

- The Desk Buddy industry contributes to the economy through the sale of both the physical device and associated services, generating revenue and taxable income.

# HARDWARE AND SOFTWARE

01

- ESP32
- DHT11 Temperature Sensor
- MQ135 Air Quality Sensor
- Light Sensor

02

- REACT (Dashboard)
- Mosquitto MQTT Broker
- Atlas(MongoDB)
- Arduino IDE

# COST PER UNIT:

# FUTURE SCOPE

01

Predicting Study Pattern using Machine Learning Models

02

Interactive LED Screen for responses to the users commands(for example touch)

03

Track if the phone is being used during the study session

04

Implement Scientific Techniques like POMODORO to make the desk buddy even more efficient